The evidence museum

Teacher guide and worked answers

Use evidence to correct museum labels about fossils, soils and light.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsb/inputs/Build SOW 2026-2027.xlsx | BUILD Weekly - Spring | C39

Review and evidence learning on fossils, soils & light (Entry Level Science / UAS).

Chronological continuation of the supplied SOW. This lesson develops or reviews the named outcome; it does not claim an accreditation award.

Prior learning

Fossils can preserve evidence of past living things.

Soils contain different materials.

Light must reach an eye for sight.

Success evidence

Use a clue from each science strand.

Correct an inaccurate exhibit label.

State one thing the evidence cannot tell us.

Equipment

Printed shared evidence and one chosen pupil route; pencil; optional ruler; projector or offline HTML.

Preparation

Print the shared evidence, one response route and exit page. Routes are alternatives, not a workload ladder.

Read the worked answers. Prepare an oral, pointing or adult-scribed route where useful.

Use this as an honest synthesis and evidence review, not a claim that these topics are all new. Optional clean sealed samples may support observation.

Practical care

No digging, tasting, crumbling fossils or handling unknown soil is required. Use printed evidence or clean sealed samples.

If using a lamp, use a cool low-power LED and keep it away from eyes. Never view the Sun.

Ready to teach alternative

All core reasoning uses the supplied evidence and can be completed in 40 minutes without a live practical.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	The museum has three doubtful labels
2	Retrieval	2-5	Retrieve one idea from each strand
3	I do	5-12	I separate a fossil from a model
4	I do	Within stage	I repair the soil label
5	I do	Within stage	I follow the museum light
6	We do	12-16	Curators compare A and B
7	We do	16-20	Repair the two-word traps
8	Hinge	20-23	Which label can stay?
9	You do	23-25	Choose your science route
10	You do	25-35	Make your own evidence record
11	You do	Within stage	Check what the evidence can show
12	Review	35-38	Lundy loop: improve the shared account
13	Review	Within stage	Keep the useful change
14	Exit	38-40	Three final checks

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening The museum has three doubtful labels

Read the objective. Pupils may edit by choosing, pointing or writing; each route uses all three strands.

2 Retrieval Retrieve one idea from each strand

Accept ancient life; mineral grains, organic matter, air, water or organisms; eyes receive light. Record uncertainty for review.

3 I do I separate a fossil from a model

Think aloud: “A has a geological history. B was made today. Similar shape does not mean the same origin.” A fossil need not preserve every feature of an organism.

4 I do I repair the soil label

Shares I do time. Think aloud: “Card C lists decayed leaf fragments. That single clue defeats the word only in ‘soil is only rock’.” The diagram is not a measured recipe.

5 I do I follow the museum light

Shares I do time. Think aloud: “The fossil is not making light. I trace both arrows to the same lit surface.” An ordinary object can be visible with reflected light.

6 We do Curators compare A and B

Prediction partners choose first, then read the card evidence. W1: A fossil; B model. Neither card reveals the ancient shell’s exact original colour.

7 We do Repair the two-word traps

W2: Soil is a mixture including mineral and organic material. Eyes receive light; the lamp is the source. Use cards C-E and diagrams. Reveal after a pupil commits to a correction.

Reveal: Use the given components and the lamp → exhibit → eye path.

8 Hinge Which label can stay?

Correct second option. Re-teach evidence versus appearance if the fossil/model distinction is insecure.

A. Today’s clay imprint is an ancient fossil. - Its known history is a model made today.

B. The fossil can be seen when reflected light reaches the eye. - This matches the light path in D.

C. Soil contains no organic material. - Card C records decayed leaf fragments.

9 You do Choose your science route

Set ONE route. Use one route with the shared records. Credit a accurate spoken museum label equally to a written label. Check that each claim has a specific clue.

10 You do Make your own evidence record

Use one route with the shared records. Credit a accurate spoken museum label equally to a written label. Check that each claim has a specific clue.

11 You do Check what the evidence can show

Shares independent time. Ask for a reason, not just a correct word. Use one route with the shared records. Credit a accurate spoken museum label equally to a written label. Check that each claim has a specific clue.

12 Review Lundy loop: improve the shared account

Listen to a selected pupil revision and visibly adopt a scientifically accurate contribution. Offer private, written or partner feedback.

Reveal: Words such as only, every or disappeared can go beyond the evidence.

13 Review Keep the useful change

Shares review time. Name how a pupil contribution changed a label, method or conclusion. Do not rank pupils.

Reveal: Words such as only, every or disappeared can go beyond the evidence.

14 Exit Three final checks

Collect brief individual evidence. Use the answer key and re-teach the specified misconception if needed.

Answers and assessment

Retrieval 1-3

1 ancient life: remains or traces. 2 relevant soil component. 3 receive light.

We do W1-W2

W1 A is the ancient fossil; B is a recent model, based on their stated histories. W2 soil is a mixture, including mineral and organic material; eyes receive light from the lamp-lit exhibit.

Hinge

Second option correct. The first ignores when B was made; the third contradicts C’s organic material.

S1-S4

S1 A. S2 any two recorded components: mineral grains, decayed leaf matter, air or water. S3 into the eye. S4 e.g. this fossil preserves evidence of an ancient shell; soil contains a mixture; reflected light reaches the eye.

M1-M4

M1 A has geological preservation; B was made today. M2 e.g. soil contains mineral grains and organic matter, with pore spaces. M3 lamp → fossil surface → eye; arrows into eye. M4 exact original colour is unknown from this record.

T1-T4

T1 some rocks contain fossils, but a stone need not preserve remains or traces of life. T2 no: the record explicitly says unknown. T3 E states that the same fossil remains but no light reaches the eye. T4 a fresh leaf is recent organic matter; ordinary falling into soil is not evidence of ancient preservation.

Exit E1-E3

E1 ancient remains or traces of life. E2 e.g. organic matter, water, air or a living organism. E3 light, here reflected from the exhibit.

R1-R2

Responses vary. Credit a scientific revision, evidence-based defence or relevant next question. A private or oral response has equal value.

Teaching and assessment notes

40 minutes: opening 2, retrieval 3, I do 7, We do 8, hinge 3, independent 12, review 3, exit 2. Zero-minute slides share the named stage time.

BUILD uses the supplied Year 3 science anchor with age-respectful contexts for older pupils. Supported, standard and stretch are selectable routes within the pathway.

Interactive We do: predict first, test against the controlled SVG model or supplied evidence, then explain or revise. Shared W tasks offer the equivalent paper discussion. Animation is a teaching representation, not filmed experimental evidence.

Keep constructed evidence separate from actual class observations. Do not invent measurements or require internet research.

Assessment: secure core is one accurate statement from each strand and use of one actual clue. Re-teach only the weak strand; keep the pupil’s correct evidence visible.

Access and responsive teaching

Supported

Short choices, word bank and pointing or adult-scribed reasons; same core scientific goal.

Standard

Make a short evidence-based explanation or record using the supplied information.

Stretch

Apply the core idea to a changed case, evaluate evidence or identify a limit.

Regulation

Preview the sequence. Offer seated observation, quiet paired rehearsal, private feedback and a pass from public sharing. Routes respond to current access needs, not a diagnosis.

Misconceptions to check

Every stone is a fossil.

A fossil is preserved evidence of past life; a stone alone need not contain this evidence.

Check: Ask what evidence of life is present.

Soil is only tiny rock.

Soil also contains organic matter and spaces with air or water.

Check: Ask which card shows a non-mineral component.

An eye lights up an exhibit.

Light enters the eye from the lit exhibit.

Check: Trace the arrow at the eye.

Word	Meaning
fossil	Preserved remains or traces of ancient life.
soil	A mixture that includes mineral and organic material.
reflect	Send arriving light back from a surface.
evidence	Information that supports or challenges an idea.

Scientific references

Department for Education: science programmes of study

Checked 8 September 2026. Year 3 plants, rocks, light and forces underpin the SOW. Accessible teaching models are adapted to the named lesson objective.

British Geological Survey: fossils

Checked 8 September 2026. Preserved ancient remains or traces; formation and limits of fossil evidence.